Quantum Algorithm Roadmap for Modular Fault-Tolerant Systems

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Abstract—Fault-tolerant quantum computing promises to solve problems in quantum chemistry, combinatorial optimisation, and cryptanalysis that remain intractable for classical hardware. However, the field suffers from a credibility gap: flagship targets such as simulating the FeMo-cofactor (FeMoco) of nitrogenase are invoked without specifying the intermediate milestones that must be cleared first. This paper closes that gap for the QONTOS modular architecture. We define a benchmark staircase that begins with classically verifiable molecules (H_2, LiH), ascends through small and medium active-space catalyst models, and culminates in the FeMoco stretch target. Each rung carries explicit resource estimates drawn from published literature, a validation gate that must be passed before the next rung is attempted, and a claim label that distinguishes demonstrated results from projections. We further present resource estimation methodology grounded in the surface-code cost model, optimisation and machine-learning workload ladders, and a scenario-based outlook linking algorithm readiness to the QONTOS 2030 hardware roadmap.

Claim label [CL-FRAMEWORK]: Algorithm roadmap paper. All resource estimates are literature-derived or projected; no original experimental data is reported.

I. Introduction

A. Motivation

Quantum algorithms are the ultimate justification for building fault-tolerant quantum hardware. Yet the path from today's noisy intermediate-scale quantum (NISQ) devices to commercially relevant problem sizes is neither smooth nor guaranteed. A responsible algorithm roadmap must (i) anchor claims in validated intermediate benchmarks, (ii) provide transparent resource estimates, and (iii) clearly label every claim along a spectrum from "demonstrated on hardware" to "projected stretch target."

The QONTOS programme addresses a hybrid superconducting-photonic modular architecture with photonically-interconnected superconducting modules, targeting logical-qubit counts in the hundreds to low thousands by the late 2020s. This paper defines the algorithmic workloads that such an architecture should aim to execute, arranged as a staircase of increasing difficulty.

B. Scope and Claim Posture

This document is a roadmap and resource-estimation paper. It does not report new experimental results. Every quantitative figure is drawn from peer-reviewed literature or from resource-estimation tools applied to published Hamiltonians. Claim labels are used throughout:

TABLE I
Claim label definitions.

Label	Meaning
CL-DEMO	Demonstrated on quantum hardware or validated simulator
CL-EST	Resource estimate derived from published methodology
CL-LIT	Claim sourced from peer-reviewed literature
CL-PROJ	QONTOS projection, not yet externally validated
CL-STRETCH	Aspirational target beyond current demonstrated capability

C. Benchmark Ladder: Design Principles

A benchmark ladder is credible only when each rung satisfies three criteria:

1) Classical verifiability at the base. The lowest rungs (H_2, LiH) must produce results that can be checked against exact classical diagonalisation or high-accuracy coupled-cluster calculations.
2) Monotonic resource growth. Qubit count, gate depth, and T-gate budget must increase predictably from one rung to the next, so that success at rung k provides evidence that the architecture can support rung $k+1$.
3) Scientific relevance at the top. The stretch target must address a problem whose solution would constitute a genuine advance in science or industry—not merely a demonstration of qubit count.

FeMoco satisfies criterion 3: accurate simulation of the FeMo-cofactor reaction mechanism would illuminate biological nitrogen fixation and potentially guide catalyst design for industrial ammonia synthesis [CL-LIT, Reiher et al. 2017 [1]].

II. Chemistry Benchmark Staircase

A. Overview

The chemistry staircase is the primary workload ladder for the QONTOS hybrid superconducting-photonic modular platform. Algorithms run on interconnected superconducting modules, with circuits partitioned across photonically-connected modules to enable distributed fault-tolerant execution. The staircase spans four stages,

Fig. 1. Application Benchmark Staircase.

each with a representative molecular system, an approximate qubit requirement, and a validation gate.

TABLE II
Chemistry benchmark staircase. Qubit counts refer to logical qubits under Jordan–Wigner or similar encoding. T-gate estimates follow the surface-code compilation model of Reiher et al. (2017) [1] and Lee et al. (2021) [2].

Stage	System	Qubits	T-gates	Status
C1	H_2, LiH, BeH_2	4–14	10^3–10^5	CL-DEMO
C2	Small catalysts	20–40	10^6–10^8	CL-EST
C3	Medium catalysts	50–100	10^9–10^{11}	CL-PROJ
C4	FeMoco-class	100–150+	$\sim 10^{14}/\sim 10^{10}$	CL-STRETCH

B. Stage C1: Classically Verifiable Small Molecules

Representative systems: H_2 (2–4 qubits in minimal basis), LiH (8–12 qubits in STO-3G to cc-pVDZ), BeH_2 (up to 14 qubits).

Purpose: These molecules are small enough that full configuration interaction (FCI) energies are available for direct comparison. They serve as the software validation harness: any quantum algorithm implementation, whether variational (VQE) or fault-tolerant (QPE), must reproduce FCI energies to chemical accuracy (1.6 mHa) before progressing.

Algorithmic approaches:

- Variational Quantum Eigensolver (VQE): Peruzzo et al. (2014) [6] demonstrated VQE on a photonic processor for HeH^+. ADAPT-VQE (Grimsley et al. 2019 [7]) offers more compact ansatze by growing the operator pool adaptively, reducing circuit depth for small systems.
- Quantum Phase Estimation (QPE): The gold-standard fault-tolerant algorithm. For H_2 in minimal basis, QPE circuits require only $O(10^3)$ T-gates after compilation, making it an ideal first target for early fault-tolerant hardware [CL-EST].

Validation gate C1: Reproduce FCI ground-state energies for H_2 and LiH to within chemical accuracy on the QONTOS digital twin or hardware prototype. Pass/fail is binary and classically verifiable.

C. Stage C2: Small Active-Space Chemistry

Representative systems: Transition-metal complexes with 20–40 active-space qubits, such as iron-nitrosyl models $[Fe(NO)]^{2+}$ and minimal Fe_2S_2 clusters.

Purpose: These systems sit at the boundary of classical tractability. Density matrix renormalisation group (DMRG) and selected-CI methods can still provide reference energies, but active-space choices begin to matter and electron correlation effects become non-trivial. Success here demonstrates that the QONTOS compiler, error correction stack, and modular interconnect can handle scientifically meaningful chemistry.

Resource estimates [CL-EST]: Using the double-factorised qubitisation approach of Lee et al. (2021) [2], a 30-qubit active space requires approximately 10^7 T-gates at target precision of 1 mHa. With a surface-code T-factory operating at ∼10 kHz logical T-gate injection rate (see Section 2.4), wall-clock time is on the order of 10^3 seconds—feasible within a single experimental run.

Validation gate C2: Reproduce reference energies (from DMRG or selected-CI) for at least two distinct transition-metal complexes. Energy errors must be below 2 mHa. Results must be obtained using fault-tolerant QPE, not variational methods alone.

D. Resource Estimation Methodology

All resource estimates in this paper follow a four-step pipeline:

1) Hamiltonian preparation. Molecular Hamiltonians are generated using standard quantum-chemistry packages (PySCF, OpenFermion) with specified active spaces and basis sets.
2) Encoding and compilation. Fermionic Hamiltonians are mapped to qubit operators via Jordan–Wigner or Bravyi–Kitaev transforms. Circuit-level compilation targets the Clifford+T gate set using state-of-the-art methods: qubitisation (Babbush et al. 2018 [10]) for block-encoding and the tensor-hypercontraction (THC) factorisation of Lee et al. (2021) [2] for reduced T-complexity.
3) Surface-code cost model. Logical T-gates are implemented via magic-state distillation. We adopt the code-distance and factory-throughput assumptions of Beverland et al. (2022) [9]: physical error rate $p = 10^{-3}$, code distance $d = 17$–27 depending on target logical error rate, and T-factory throughput scaling as described in their resource-estimation framework. Physical-qubit overhead per logical qubit ranges from ∼3,000 (d=17) to ∼7,000 (d=27).
4) Wall-clock projection. Total runtime is estimated as (T-gate count) / (T-factory throughput × number of

parallel factories). For the QONTOS modular architecture, we assume 4–16 parallel T-factories distributed across interconnected modules [CL-PROJ].

Validation of methodology: At the C1 level, our resource-estimation pipeline reproduces the T-gate counts published by Reiher et al. (2017) [1] for FeMoco to within a factor of 2, and matches the improved estimates of Lee et al. (2021) [2] for the same system under THC factorisation.

E. Validation Gate: Progressive Chemistry Milestones

The following milestone table codifies the pass/fail criteria at each stage. The QONTOS algorithm team may not publicly claim readiness for stage $k+1$ until stage k has been cleared.

TABLE III
Progressive chemistry validation gates.

ID	Gate Criterion	Evidence Required	Label
VC-1	H_2 + LiH to chemical accuracy	Energy vs. FCI	CL-DEMO
VC-2	Two TM complexes, <2 mHa	Energy vs. DMRG/selected-CI	CL-DEMO
VC-3	Medium catalyst, <3 mHa	Energy vs. best classical	CL-DEMO
VC-4	FeMoco active-space energy	Comparison to Reiher et al. ref.	CL-DEMO
VR-1	Resource pipeline validated at C1	Reproduce published T-gate counts	CL-EST
VR-2	Resource pipeline validated at C2	Independent audit	CL-EST

F. Stage C3: Medium Catalyst Models

Representative systems: Cytochrome P450 active-site models (Goings et al. 2022 [3]), $[Fe_4S_4]$ cubane clusters, and manganese-oxo complexes relevant to water splitting.

Purpose: These systems are classically challenging. Goings et al. (2022) [3] showed that even for cytochrome P450, reliable classical electronic-structure calculations require careful multireference treatment and remain subject to methodological uncertainty. Quantum advantage in this regime is not about speed but about reliability: a fault-tolerant QPE calculation can provide a definitive energy that resolves discrepancies between competing classical methods.

Resource estimates [CL-PROJ]: A 60–80 qubit active space under THC qubitisation requires approximately 10^{10} T-gates (Lee et al. 2021 [2], extrapolated). With 8 parallel T-factories at 10 kHz, wall-clock time is of order 10^5 seconds (∼1 day). This is aggressive but potentially feasible on a QONTOS-class modular architecture with 3,000–5,000 physical qubits per module and 4–8 interconnected modules.

Validation gate C3: Obtain ground-state energies for at least one medium catalyst model with error below 3 mHa relative to the best available classical reference.

G. Stage C4: FeMoco Stretch Target

Representative system: The full FeMo-cofactor, Fe_7MoS_9C, with a (54e, 54o) or larger active space as specified by Reiher et al. (2017) [1].

Purpose: FeMoco is the iconic stretch target for fault-tolerant quantum chemistry. Its scientific importance lies in elucidating the mechanism of biological nitrogen fixation. Industrially, understanding this mechanism could guide design of ambient-condition catalysts for ammonia synthesis.

Resource estimates [CL-LIT]: Reiher et al. (2017) [1] estimated $\sim 10^{14}$ Toffoli (or equivalently T-) gates for a single QPE run on FeMoco with a (54e, 54o) active space using a first-quantised approach. Lee et al. (2021) [2] reduced this by roughly three orders of magnitude via tensor hypercontraction, bringing the estimate to $\sim 10^{10}$–10^{11} T-gates for a comparable active space—though the precise savings depend on the target precision and the THC rank. Babbush et al. (2018) [10] provided earlier improvements through low-rank encodings of electronic spectra.

Physical resource projection [CL-STRETCH]: Under the Beverland et al. (2022) [9] surface-code model with $p = 10^{-3}$ and the Lee et al. (2021) [2] T-gate counts:

TABLE IV
FeMoco physical resource estimates. These are projections, not demonstrated capabilities.

Parameter	Estimate	Source
Logical qubits	∼100–150	[1]; [2]
T-gate count (THC)	$\sim 10^{10}$–10^{11}	[2]
Code distance	21–27	[9] (extrap.)
Phys. qubits / logical	∼4,000–7,000	[9]
Total physical qubits	∼400,000–1,000,000	CL-STRETCH
T-factories (parallel)	16–32	CL-PROJ
Wall-clock (optimistic)	∼1–10 days	CL-STRETCH

Validation gate C4: This gate is aspirational. It requires: (a) successful completion of gates VC-1 through VC-3; (b) a peer-reviewed resource estimate specific to the QONTOS modular architecture; (c) demonstration that the modular interconnect supports the required logical-qubit communication patterns.

III. Optimisation Benchmark Staircase

A. Overview

Combinatorial optimisation is a second major application domain. The QONTOS optimisation ladder targets problems where quantum approximate optimisation or quantum annealing-inspired methods may offer advantage.

TABLE V
Optimisation benchmark staircase.

Stage	Workload	Qubits	Approach	Status
O1	Small QAOA	10–30	QAOA p=1..5	CL-EST
O2	Medium portfolio/logistics	50–200	QAOA, Grover-mixer	CL-PROJ
O3	Large distributed opt.	200+	Modular QAOA, hybrid	CL-STRETCH

TABLE VI
Machine learning benchmark staircase.

Stage	Workload	Qubits	Status
M1	Small feature-map and kernel tasks	4–20	CL-EST
M2	Modular QML experiments	20–100	CL-PROJ
M3	Large quantum-enhanced model targets	100+	CL-STRETCH

B. Stage O1: Small QAOA Instances

Algorithmic basis: The Quantum Approximate Optimisation Algorithm (QAOA) of Farhi et al. (2014) [5] provides a variational framework for combinatorial problems. At low circuit depth ($p = 1$–3), QAOA circuits are executable on near-term hardware and serve the same software-validation role as C1 chemistry benchmarks.

Validation gate O1: Demonstrate QAOA on MaxCut instances with 10–20 nodes, achieving approximation ratios matching or exceeding the Goemans–Williamson bound (0.878) on the QONTOS digital twin.

C. Stage O2: Medium Optimisation Problems

Representative problems: Portfolio optimisation with 50–200 assets, vehicle routing with time windows, and logistics scheduling.

Resource estimates [CL-PROJ]: For a 100-variable QAOA instance at depth $p = 10$, the circuit requires approximately 10^4 two-qubit gates (before error correction). Under surface-code compilation with $d = 11$–15, this translates to $\sim 10^7$ physical gate operations across 50–100 logical qubits.

Validation gate O2: Achieve solution quality within 5% of the best known classical heuristic on at least two distinct combinatorial problem classes.

D. Stage O3: Large Distributed Optimisation

This is the stretch target for optimisation. It requires the modular architecture to distribute QAOA or hybrid quantum-classical iterations across multiple interconnected modules. Practical advantage here would likely depend on problem-specific structure rather than generic speedup.

Claim label [CL-STRETCH]: No concrete resource estimates are provided for O3. Its inclusion signals architectural ambition, not algorithmic readiness.

IV. Machine Learning Benchmark Staircase

A. Overview

Quantum machine learning (QML) remains the most speculative of the three workload families. The QONTOS ML ladder is included for completeness but is explicitly labelled as lower maturity than the chemistry and optimisation ladders.

Claim label [CL-PROJ]: The QML ladder is a research path. No claims of quantum advantage in machine learning are made or implied.

B. Validation Gates for QML

TABLE VII
QML validation gates.

Gate	Criterion
VM-1	Reproduce published QML benchmark results on QONTOS digital twin
VM-2	Demonstrate modular execution of a QML circuit across two or more QONTOS modules
VM-3	Identify at least one dataset/task where quantum-enhanced features provide statistically significant improvement over classical baselines

V. Cryptanalytic Workload: Shor's Algorithm and RSA

A. Context

While not a primary target for the QONTOS programme, the factoring problem provides a well-understood algorithmic benchmark with precisely characterised resource requirements.

Reference estimate [CL-LIT]: Gidney and Ekera (2021) [8] showed that a 2048-bit RSA integer can be factored in approximately 8 hours using 20 million noisy qubits (at physical error rate 10^{-3}) with a highly optimised implementation of Shor's algorithm. This requires approximately 2×10^{12} Toffoli gates and a code distance of ~ 27.

QONTOS relevance: The Gidney–Ekera estimate provides an independent calibration point for the QONTOS resource-estimation pipeline. If our pipeline produces consistent estimates for this well-characterised problem, it strengthens confidence in our chemistry projections.

Claim label [CL-LIT]: QONTOS does not target cryptanalysis. This workload is included solely as a resource-estimation calibration reference.

VI. Scenario-Based Algorithm Outlook

A. Three Scenarios

The algorithm roadmap is tied to three hardware scenarios reflecting the QONTOS 2030 programme timeline.

TABLE VIII
Scenario–algorithm mapping.

Scenario	Hardware	Algorithmic Capability	Timeline
Base	1–2 modules, ∼1,000 phys. qubits each, d=7–11	C1 + O1; digital-twin validation of C2	2025–2027
Aggressive	4–8 modules, ∼3,000 phys. qubits each, d=13–17	C2 demonstrated; C3 estimated; O2 demonstrated	2027–2029
Stretch	8–16+ modules, ∼5,000+ phys. qubits each, d=21–27	C3 demonstrated; C4 estimated; O3 explored	2029–2031

B. Base Scenario Detail

In the base scenario, the QONTOS platform operates 1–2 modules with a total of ∼2,000 physical qubits. At code distance $d = 7$–11, the system supports 10–20 logical qubits with logical error rates of $\sim 10^{-4}$–10^{-6}. This is sufficient for:

- C1 benchmarks: QPE on H_2 and LiH, verifying energy to chemical accuracy.
- O1 benchmarks: QAOA at $p = 1$–3 on MaxCut instances up to 15 nodes.
- Digital-twin validation: Run C2-scale resource estimates to confirm pipeline accuracy before hardware is available.

Claim label [CL-PROJ]: The base scenario is a projection based on the current QONTOS hardware development timeline.

C. Aggressive Scenario Detail

In the aggressive scenario, 4–8 interconnected modules provide 12,000–24,000 physical qubits. At $d = 13$–17, the system supports 40–80 logical qubits. This enables:

- C2 demonstration: QPE on small transition-metal complexes.
- O2 demonstration: QAOA on medium optimisation instances.
- C3 resource estimation: Detailed cost projections for P450 and Fe_4S_4 models.

The modular interconnect must support inter-module logical gates at rates compatible with the C2 circuit requirements. Cross-module entanglement via photonic interconnects is assumed (see QONTOS Paper 04).

D. Stretch Scenario Detail

The stretch scenario requires 8–16+ modules with total physical-qubit counts approaching 50,000–100,000. This is the minimum envelope for C3 demonstration and preliminary C4 resource estimation with architecture-specific parameters.

Kim et al. (2023) [4] demonstrated that quantum hardware can produce expectation values beyond brute-force classical simulation for certain structured problems on 127 qubits. While this does not constitute fault-tolerant quantum advantage, it illustrates that the gap between NISQ utility demonstrations and early fault-tolerant benchmarks is narrowing. The stretch scenario positions QONTOS at the frontier of this transition.

VII. Algorithm–Architecture Co-Design Considerations

A. Modular Connectivity Requirements

Chemistry algorithms (especially QPE with qubitised Hamiltonians) require all-to-all logical qubit connectivity within the active space. In a modular architecture, this translates to inter-module logical gate requirements that scale with the number of active-space qubits distributed across modules.

Key design constraint [CL-PROJ]: For C2 workloads, inter-module gate rates of ∼1 kHz are sufficient. For C3 workloads, rates of ∼10 kHz are required. For C4, rates approaching ∼100 kHz may be needed, depending on circuit parallelism.

B. T-Factory Distribution

The dominant resource bottleneck in fault-tolerant chemistry is T-gate throughput. The QONTOS modular architecture naturally supports distribution of T-factories across modules:

- Base: 1–2 T-factories, throughput ∼1 kHz aggregate
- Aggressive: 4–8 T-factories, throughput ∼10 kHz aggregate
- Stretch: 16–32 T-factories, throughput ∼100 kHz aggregate

These throughput targets assume the T-factory designs analysed in Beverland et al. (2022) [9] and are consistent with the surface-code parameters in Section 2.4.

C. Compiler and Scheduling Requirements

The QONTOS software stack (Paper 06) must support:

1) Automatic active-space selection for chemistry Hamiltonians.
2) Hamiltonian encoding via qubitisation or THC factorisation.
3) Surface-code circuit compilation with T-gate scheduling.
4) Cross-module gate routing for modular execution.

These requirements are derived from the algorithm staircase; the compiler team should prioritise C1 and C2 circuit compilation first, then extend to C3 patterns.

VIII. Comparison to Industry Resource Estimates

A. Literature Landscape

Several independent groups have published resource estimates for fault-tolerant quantum chemistry:

TABLE IX
Comparison of published resource estimates.

Source	Target	Log. Q.	T-gates	Phys. Q.	Label
Reiher '17	FeMoco	∼108	$\sim 10^{14}$	$\sim 10^{6}$	CL-LIT
Lee '21	FeMoco (THC)	∼350	$\sim 10^{10}$	$\sim 4\times 10^{5}$	CL-LIT
Babbush '18	Elec. spectra	Varies	Linear T	Varies	CL-LIT
Beverland '22	Framework	Varies	Varies	Framework	CL-LIT
Gidney '21	RSA-2048	∼3–4k	$\sim 2\times 10^{12}$	$\sim 20\times 10^{6}$	CL-LIT
Goings '22	P450 (class.)	N/A	N/A	N/A	CL-LIT

B. QONTOS Positioning

The QONTOS resource-estimation pipeline does not claim to improve upon these published figures. Instead, it translates them into architecture-specific costs for the modular superconducting platform:

- Module count and physical-qubit allocation per module
- Inter-module entanglement budget
- T-factory placement and scheduling
- Wall-clock time under realistic duty-cycle assumptions

Claim label [CL-PROJ]: Architecture-specific cost translation is a QONTOS contribution, but the underlying algorithmic costs are taken directly from published literature.

IX. Risk Factors and Honest Unknowns

A. Chemistry Risks

1) Active-space inadequacy. FeMoco may require active spaces larger than (54e, 54o) to capture all relevant correlation effects. This would increase resource requirements beyond current estimates.
2) Basis-set effects. Moving from minimal to larger basis sets (e.g. cc-pVTZ) increases qubit counts and T-gate budgets significantly.
3) Excited-state complexity. Reaction mechanism elucidation requires not just ground-state energies but energy differences along reaction coordinates, which may require multiple QPE runs per data point.

B. Architecture Risks

1) Inter-module gate fidelity. If photonic interconnect fidelity falls below the surface-code threshold, modular execution of C3+ workloads may not be feasible.
2) T-factory throughput. Achieving 10+ kHz aggregate T-gate throughput requires engineering advances beyond current demonstrations.
3) Decoder latency. Real-time decoding must keep pace with syndrome extraction rates; backlog would stall computation.

C. Algorithmic Risks

1) Classical competition. Advances in tensor-network, selected-CI, and other classical methods may shift the quantum-advantage boundary upward. Goings et al. (2022) [3] demonstrated that even P450, once thought to require quantum methods, can be treated reliably with careful classical approaches.
2) Error mitigation vs. error correction. The transition from NISQ error-mitigated results (as in Kim et al. 2023 [4]) to fully fault-tolerant execution involves a discontinuity in resource requirements that may be larger than currently projected.

X. Conclusion

The QONTOS algorithm roadmap is built on a benchmark staircase, not a single flagship claim. The staircase begins with small molecules (H_2, LiH) that are classically verifiable, progresses through transition-metal complexes and medium catalyst models that test the boundaries of classical methods, and culminates in the FeMoco stretch target that would represent a genuine scientific advance.

Every rung of the staircase carries:

- A resource estimate grounded in published methodology (Reiher et al. 2017 [1], Lee et al. 2021 [2], Beverland et al. 2022 [9]).
- A validation gate that must be cleared before the next rung is attempted.
- A claim label that distinguishes demonstrated results from projections.

This structure ensures that the QONTOS algorithm narrative is both ambitious and honest. FeMoco remains the north star, but the programme's credibility is built rung by rung, not by invoking the destination alone.

Summary claim posture:

TABLE X
Summary claim posture.

Assertion	Label
QONTOS has defined a benchmark staircase from H_2 to FeMoco	CL-FRAMEWORK
C1 benchmarks are within reach of early FT hardware	CL-EST
C2 benchmarks require 4–8 module systems at d=13–17	CL-PROJ
C3 benchmarks require engineering advances	CL-PROJ
C4 (FeMoco) is a stretch target requiring $\sim 10^{10}$–10^{14} T-gates	CL-STRETCH
Resource estimates are derived from Reiher, Lee, and Beverland	CL-LIT
No demonstrated quantum advantage is claimed	Honest ack.

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